

Android was first developed 2003 by the company Android Inc which was later bought by Google in 2005. What followed were multiple breakthroughs one after another finally resulting in the massive giant of an OS that we see today.

Android is no longer limited to phones, today wearables, smart TVs and even automobile navigation and entertainment units are running Android. This allows unprecedented access to developers in terms of creativity and poses an equally big challenge - how do we master the seemingly limitless ecosystem of Android development?

Short answer, we don't.

We can however, learn in a reactive manner the areas of Android programming that we need. For example, someone designing a mobile app for a banking firm would require little to no knowledge of the Android automobile APIs and hence could delay learning about it unless required at a later date.

To begin programming in Android, we need our 'development environment' set up in the correct way, to ensure rapid prototyping and testing. For this we use the Android Studio, as recommended on the Google developers website. It provides a complete development environment for programmers to start writing basic apps for the Android OS.

Additionally the Android NDK or Native Development Toolkit allows developers to harness the powers of native programming, if they feel that the Android libraries won't do the job. For most applications, however, Android Studio and SDK Tools are enough. We can download both the Android Studio and SDK Tools as well as Android NDK from the Google developers page. Once downloaded, simply run the Android Studio and you are ready to develop your first Android app!

Anatomy of a Basic Android App

A basic Android application consists of a collection of at least one these components (among others):

- **Activity** - An activity is a fixed unit of work that can be triggered at any point by an external source. Activities are the most basic building blocks of Android programming and almost all applications have at least one activity defined within them.
- **Intent** - An intent is the requirement of an action to be performed. It is subtly different from an activity in the sense that an activity is a unit of work but an intent is a directive to perform said work. Intents are what